

8.3.1.2.2 Reduced Test Time

A special debug and test configuration bit is provided in the SYS_CTRL2 register, called *[DELAY_DIS]*. Setting *[DELAY_DIS]* bypasses the OV/UV protection fault timers and allows a fault condition to be registered within 200 ms after application of such a fault condition.

8.3.1.3 Control Subsystem

8.3.1.3.1 FET Driving (CHG AND DSG)

Each BQ769x0 device provides two low-side FET drivers, CHG and DSG, which control NCH power FETs or may be used as a signal to enable various other circuits such as a high-side NCH charge pump circuit.

Both DSG and CHG drivers have a fast pull-up to nominally 12 V when enabled. DSG uses a fast pull-down to VSS when disabled, while CHG utilizes a high impedance (nominally 1 M Ω) pull-down path when disabled.

An additional internal clamp circuit ensures that the CHG pin does not exceed a maximum of 20 V.

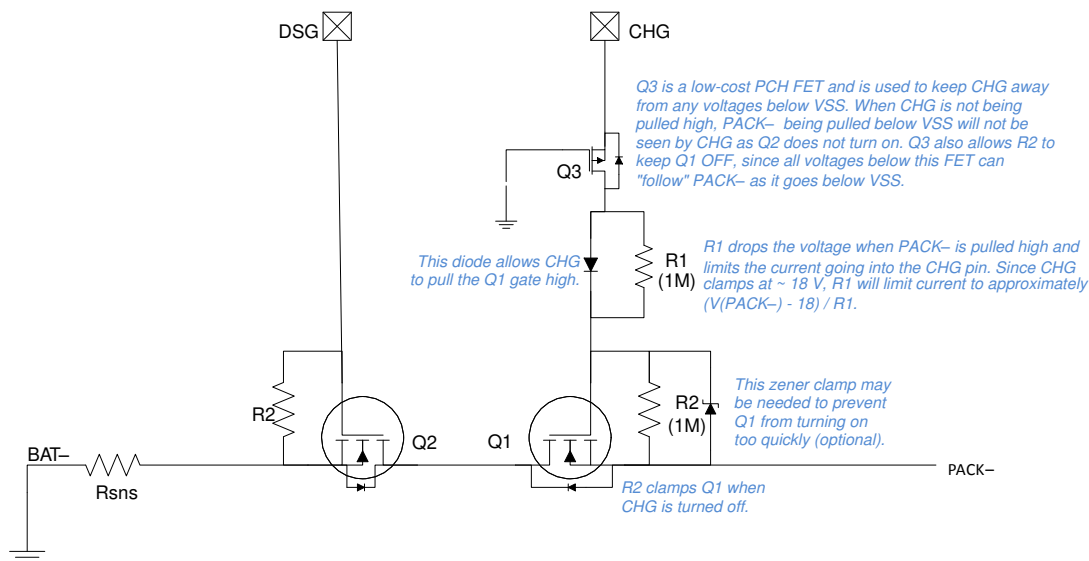


图 8-2. CHG and DSG FET Circuit

The power path for the CHG and DSG pull-up circuit originates from the REGSRC pin, instead of BAT.

To enable the CHG fet, set the *[CHG_ON]* register bit to 1; to disable, set *[CHG_ON]* = 0. The discharge FET may be similarly controlled through the *[DSG_ON]* register bit.

Certain fault conditions or power state transitions will clear the state of the CHG/DSG FET controls. 表 8-1 shows what action, if any, to take to *[CHG_ON]* and *[DSG_ON]* in response to various system events:

8.3.1.3.1.1 High-Side FET Driving

The BQ769x0 battery monitors provide low-side FET drivers that work well for many systems. For some systems, high-side FETs may also be beneficial. High-side FETs enable continuous communication between a host controller and the monitor, regardless of whether the FETs are on or off. This allows the controller to read critical pack parameters despite safety faults, enabling the system to access pack conditions before allowing normal operations to resume. The BQ769200 high-side N-channel FET driver can be used with the BQ769x0 monitor in systems where high-side FETs are needed. See 图 9-4.

表 8-1. CHG, DSG Response Under Various System Events

EVENT	[CHG_ON]	[DSG_ON]
OV Fault	Set to 0	—
UV Fault	—	Set to 0
OCD Fault	—	Set to 0